

Material requirements of global electricity sector pathways to 2050 and associated greenhouse gas emissions

– Supplementary material –

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1 Electricity infrastructures in 2050

1.1 Power plants

The figures S1 and S2 illustrate the ranges of installed power plant capacities in 2050 in the 281 considered IAMC scenarios. In Figure S1, the scenarios are grouped by the categories of global temperature rise and in Figure S2 by the underlying Shared Socioeconomic Pathways (SSPs; see Riahi et al., 2017). The installed capacities in 2017 (see Kalt et al., 2021) are shown for comparison.

Exceptionally high solar capacity expansion occurs in many scenarios developed with the model/modelling framework REMIND/REMIND-MAGPIE (Bertram et al., 2018b; Kriegler et al., 2018; Strefler et al., 2018). Figure S2 shows that scenarios with high solar power expansion are primarily based on SSP1 and SSP2, however, those with the highest solar power capacities in 2050 are not based on a specific SSP. The continued reliance on fossil fuels in SSP5 and SSP3, obvious from high gas and coal capacities, is part of the narratives of these pathways (see Table S1).

Table S1. Narratives of the Shared Socioeconomic Pathways and characteristics regarding technological development (Bauer et al., 2017; Riahi et al., 2017).

No.	Description	Characteristics (Bauer et al., 2017)
SSP1	<i>Sustainability – Taking the Green Road</i>	Technological change directed away from fossil fuels, towards efficiency and renewables; low carbon and energy intensity
SSP2	<i>Middle of the Road</i>	Some investments in renewables but continued reliance on fossil fuels; medium carbon intensity; uneven energy intensity
SSP3	<i>Regional Rivalry – A Rocky Road</i>	Slow technological change, directed towards domestic energy sources; high energy and carbon intensity in regions with large domestic fossil fuel resources
SSP4	<i>Inequality – A Road Divided</i>	Diversified investments including efficiency and low-carbon sources; low/medium carbon and energy intensity
SSP5	<i>Fossil-fueled Development</i>	Directed towards fossil fuels; alternative sources not actively pursued; high carbon intensity

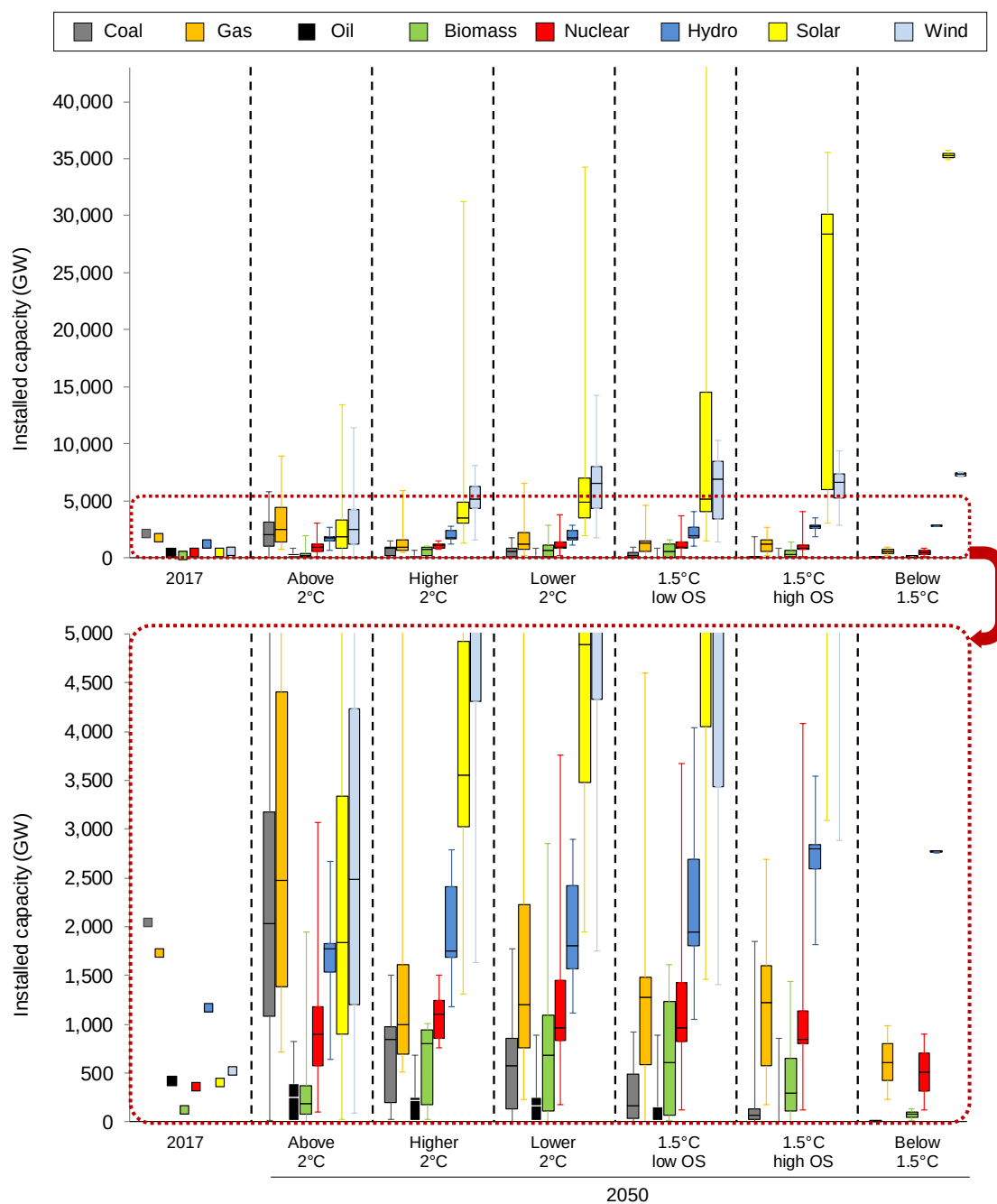


Figure S1. Installed capacities of power generation technologies in 2017 and ranges in 2050 according to scenarios with different levels of temperature rise (data from Huppmann et al., 2019; evaluation and illustration by the authors; OS = temperature overshoot).

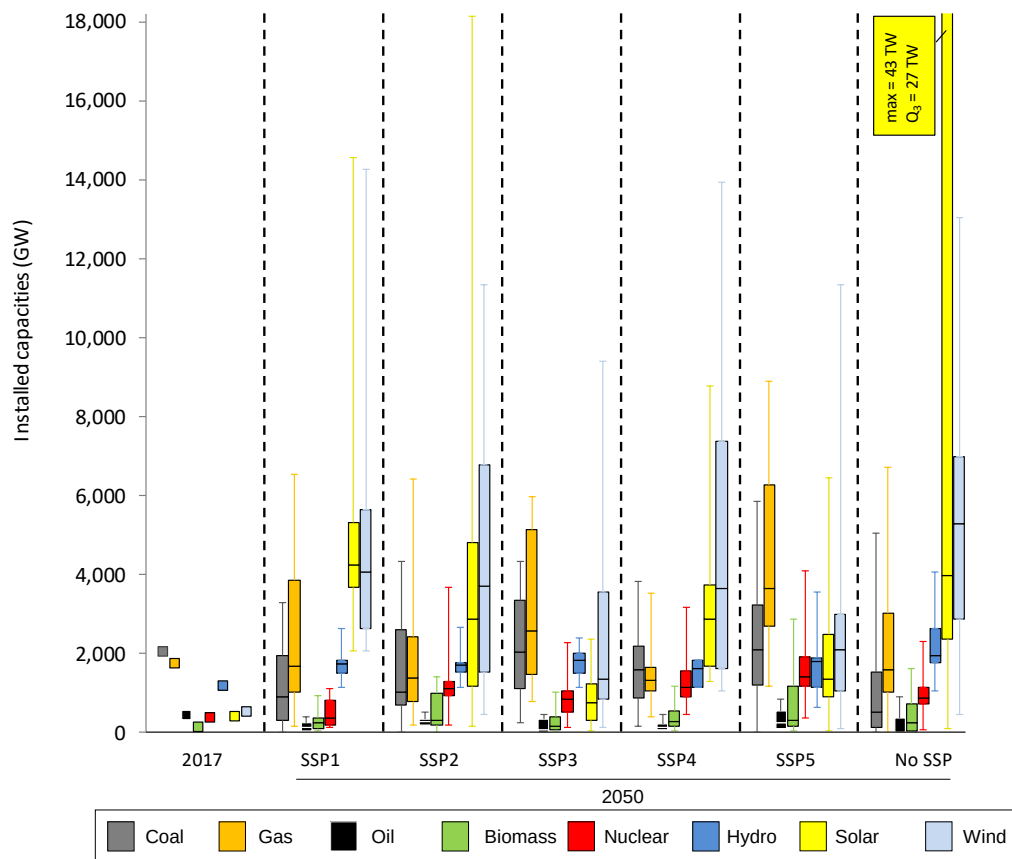


Figure S2. Installed capacities of power generation technologies in 2017 and ranges in 2050 according to SSP scenario ensemble. Scenarios categorized as “No SSP” partly also draw from specific SSP narratives (e.g. Bertram et al., 2018a; Kriegler et al., 2018; Strefler et al., 2018), but are not part of the SSP scenario ensemble (Bauer et al., 2017; Rogelj et al., 2018) (data from Huppmann et al., 2019; evaluation and illustration by the authors).

1.2 Grids and transformers

Grid lengths are calculated with the regression models presented in Kalt et al. (2021a). Transmission grid lengths are derived from electricity generation capacities (obtained from the IAMC database, Huppmann et al., 2019), land area (UNSD, 2016) and population with access to electricity, and distribution grids from electricity use (IAMC database) and the number of households with access to electricity. Projections for the population and number of households with access to electricity are based on scenario-specific population developments (IAMC database) and self-developed projections for the share of population with access to electricity and average household sizes (see Figure S3). The current share of population with access to electricity in the five world regions is calculated from country-level data according to the World Bank (2020) and assumed to increase to 100 % until 2050. Values for intermediate years are simplistically derived through linear interpolation. Average household sizes in 1980 and 2017 are obtained from UNDESA (2019). Historical trends are linearly extrapolated to 2050.

Capacities of transmission and distribution transformers (in Gigavolt-Ampere) are also estimated according to the approach presented in Kalt et al. (2021a). That is, we assume transformer-to-generator ratios of 2.5 for transmission transformers as well as distribution transformers. Generator step-up transformers are integral parts of power plants and usually within the system boundaries of

power plant LCAs. This transformer type is thus not considered as individual component in the material stock assessment (cf. Kalt et al., 2021b).

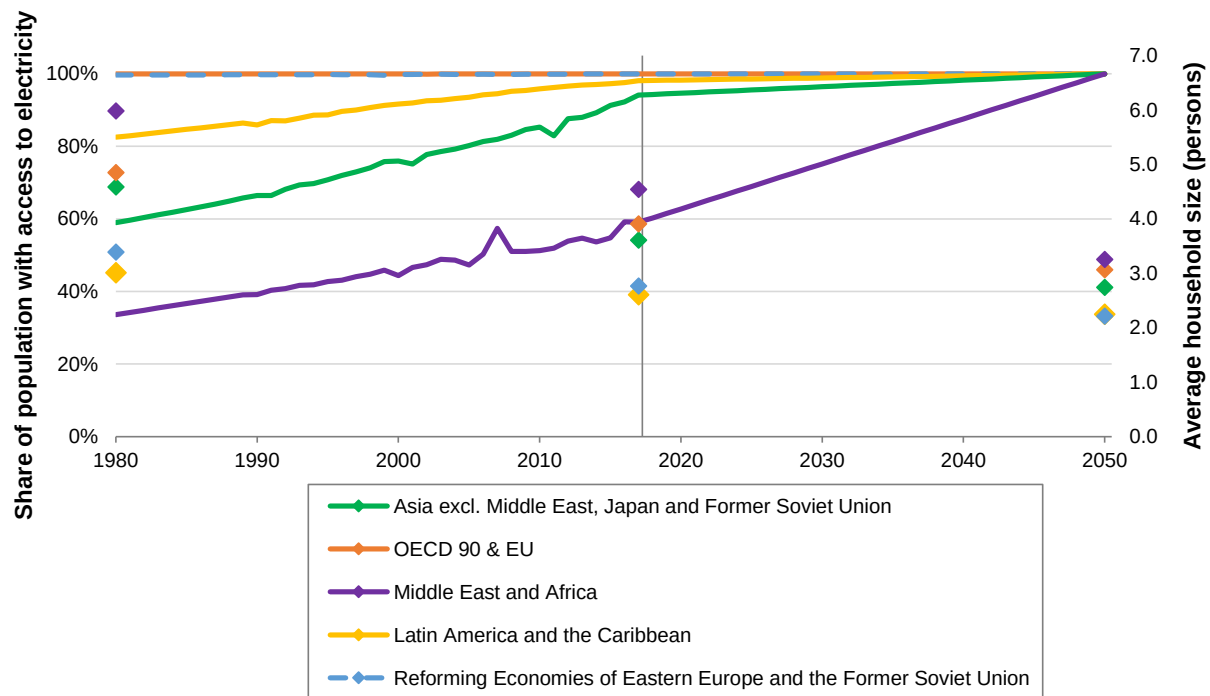


Figure S3. Assumed projections for the shares of population with access to electricity (lines) and average household sizes (diamond-shaped markers). Historical data are obtained from World Bank (2020) and UNDESA (2019), respectively. Country groupings correspond to those used in the IAMC database (Huppmann et al., 2019).

In absence of universal definitions of transmission and distribution grids, Kalt et al. (2021a) classify overhead lines and underground cables with a voltage level up to 100 kV as distribution and such with more than 100 kV as transmission grid. The following figures illustrate the global circuit lengths of both categories in 2017 and 2050. Circuit lengths refer to the combined length of all circuits (corresponding to three conductors in three-phase AC systems and two conductors in DC systems). Distances covered by multiple-circuit lines are thus multiplied with the number of circuits to obtain circuit lengths.

In scenarios with largely fossil fuel-based power generation (Above 2°C; SSP3; SSP5), the average/median estimate for the global transmission circuit length is in the range of about 6 to 8 million kilometres in 2050, corresponding to less than a doubling relative to 2017. For scenarios with electricity generation largely based on renewables, our results indicate that an increase beyond 100 % or even beyond 200 % is likely. In contrast, the global distribution grid length increases by about 100 % in all scenario groups, mainly driven by grid expansion in the regions “Middle East and Africa” and less developed Asian countries (Figure S6).

Since the regression used for deriving these estimates is based on empirical data from existing (centralized) power systems, our approach does not consider the possible influence of technological innovation in grid structures (i.e. decentralized grids based on distributed generation; smart grids). In case of large-scale deployment of such innovations, future transmission grids in largely renewable-based electricity systems could be significantly less extensive than estimated here. On the other hand, large-scale deployment of decentralized power generation (especially solar PV) is likely to make grid

reinforcement necessary in existing distribution grids; this aspect is also disregarded here. In other words, massive deployment of distributed generation is considered likely to shift material requirements from transmission to distribution grids. To what extent this may influence the overall material requirements of electricity grids remains to be investigated in subsequent studies, based on grid simulation models.

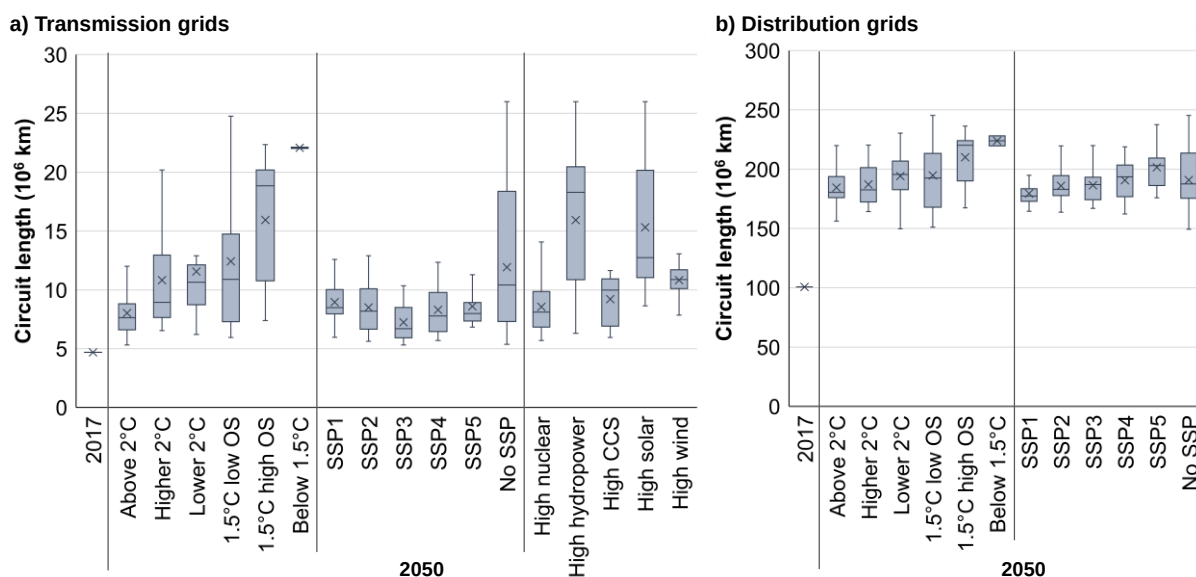


Figure S4. Transmission (a) and distribution (b) grid lengths in 2017 and projections to 2050. No capacity-based scenario groupings are shown in panel b because the regression for distribution grids is independent from power plant capacities.

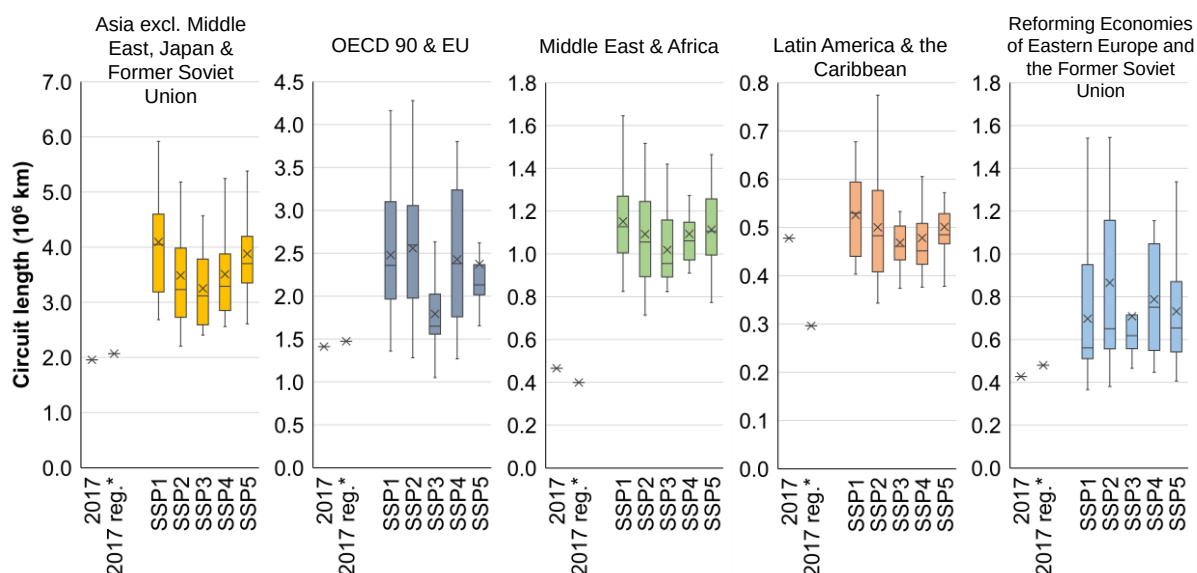


Figure S5. Estimates for transmission grid lengths in IAMC scenarios ('2017 reg.' indicates circuit lengths according to regression results, '2017' the lengths derived from GIS data, national and international statistics on transmission grids (Kalt et al., 2021a), which are shown for comparison. Estimates of material stocks and flows are entirely based on regression results.

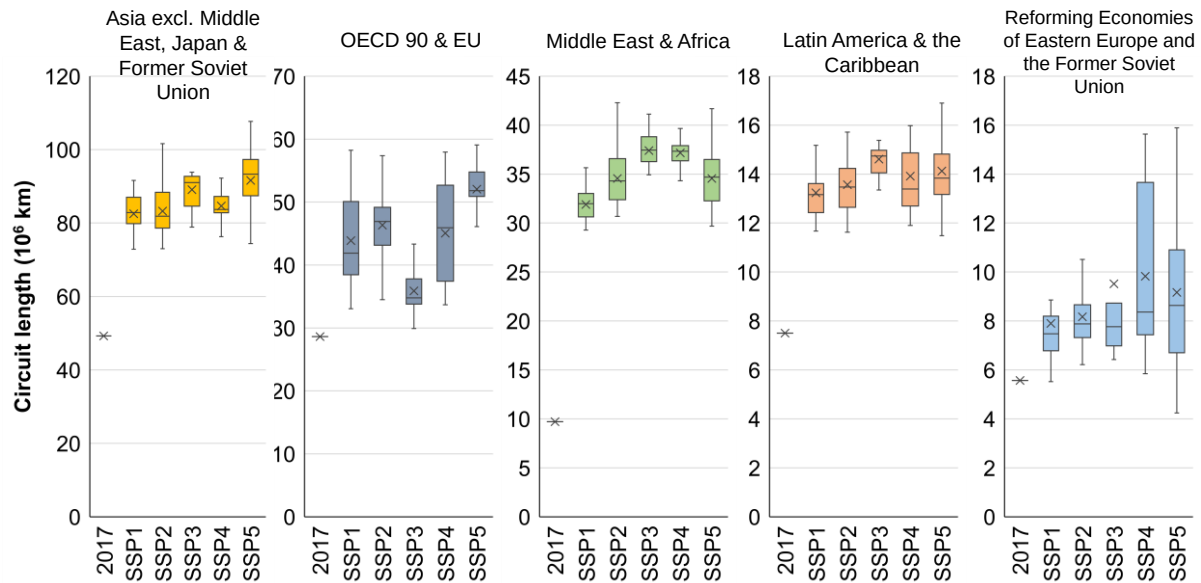


Figure S6. Estimates for distribution grid lengths in IAMC scenarios

2 Material intensities and lifetimes

2.1 Power plants

Table S2 summarizes the material intensity ranges assumed for electricity generation technologies. Graphical illustrations are provided in Figure S7. For the literature used to derive these ranges, please refer to Kalt et al. (2021b). For power plants with carbon capture and storage (CCS), the existing literature indicates that bulk material requirements are up to twice as high as those of the corresponding conventional power plants (without CCS). Based on Singh et al. (2015), Volkart et al. (2013), Luderer et al. (2019) and Cormos and Agachi (2012), we assume that the material intensities, including infrastructure for the transport and injection of CO₂, are at least 50 % and up to 100 % higher than those of the conventional power plants without CCS.

For the Monte Carlo simulation in section 3.1 of the main article, random material intensity values within the stated ranges (equal distribution) are determined, individually for each world region. For the low, medium and high estimates in section 3.2, we assume the lower bounds, the medium estimates and upper bounds stated in Table S2, respectively.

Differences in the material intensities of different solar and wind sub-technologies (i.e., PV and concentrating solar power, and onshore and offshore wind turbines) are taken into account if capacity data are differentiated accordingly. However, not all scenarios in the IAMC database provide differentiations between sub-technologies. If no differentiation is available, we assume the average shares from all scenarios with differentiation (7% CSP and 93 % PV, and 73% onshore and 27% offshore, respectively).

The material intensities for “material-efficient” solar technologies are based on the following assumptions: First, only PV is assumed if there is no differentiation between PV and CSP in the considered scenario. Second, with regard to the aluminium intensity of PV, we narrowed the range down to ranges derived from more recent studies characterized by relatively low aluminium intensities; this leads to an upper bound corresponding to less than 50 % of the upper bound of the

default range. Third, no concrete demand is assumed in the ‘material-efficient case’ for solar power, because building-integrated solar PV is assumed.

For wind, we excluded the highest data for iron/steel found in literature. This leads to an about 50%-reduction of the upper bound, compared to the default case. For offshore turbines, only foundation types without concrete are assumed in calculations of the “material-efficient” case.

Table S2. Material intensity ranges for power plants (based on Kalt et al., 2021b; assumptions on material intensities of power plants with CCS are explained in the text)

Unit: g/W			Biomass without CCS	Biomass with CCS	Coal without CCS	Coal with CCS	Gas without CCS	Gas with CCS	Geo-thermal	Hydro	Nuclear	Oil	Solar	Solar CSP	Solar PV	Wind	Wind Offshore	Wind Onshore
Iron, steel	Default range	lower bound	70,00	105,00	30,00	45,00	10,00	15,00	15,00	20,00	70,00	15,00	19,34	170,00	8,00	115,12	156,00	100,00
		medium estimate	197,50	377,50	65,00	122,50	30,00	57,50	618,84	110,00	90,00	32,50	121,65	510,00	92,42	233,13	478,73	142,29
		upper bound	325,00	650,00	100,00	200,00	50,00	100,00	1.222,67	200,00	110,00	50,00	223,97	850,00	176,85	351,14	801,47	184,58
	Range for 'material-efficient techs/designs'	lower bound																
		medium estimate																
		upper bound																
Aluminium	Default range	lower bound	1,00	1,50	0,20	0,30	0,10	0,15	3,80	0,27	0,20	0,65	12,27	2,60	13,00	0,78	0,78	0,78
		medium estimate	2,25	4,25	1,60	3,15	0,85	1,68	5,30	0,57	0,60	1,00	30,23	13,30	31,50	2,68	2,68	2,68
		upper bound	3,50	7,00	3,00	6,00	1,60	3,20	6,80	0,88	1,00	1,35	48,18	24,00	50,00	4,58	4,58	4,58
	Range for 'material-efficient techs/designs'	lower bound																
		medium estimate																
		upper bound																
Copper	Default range	lower bound	1,50	2,25	0,70	1,05	0,60	0,90	1,00	0,50	1,40	0,65	0,99	0,80	1,00	1,51	2,00	1,33
		medium estimate	2,50	4,63	1,95	3,73	1,10	2,05	1,70	3,69	8,20	0,93	3,21	3,90	3,16	4,58	7,11	3,64
		upper bound	3,50	7,00	3,20	6,40	1,60	3,20	2,40	6,88	15,00	1,20	5,44	7,00	5,32	7,65	12,23	5,95
	Default range	lower bound	380,00	570,00	100,00	150,00	45,00	67,50	120	2,500	400	50,00	7,00	100,00	0,00	185,42	0,00	254,00
		medium estimate	540,00	985,00	275,00	525,00	60,00	108,75	120	5,250	650	62,50	350,93	465,00	342,35	580,03	791,00	502,00
		upper bound	700,00	1.400,00	450,00	900,00	75,00	150,00	120	8,000	900	75,00	694,87	830,00	684,70	974,64	1.582,00	750,00
Concrete	Range for 'material-efficient techs/designs'	lower bound																
		medium estimate																
		upper bound																
	Default range	lower bound																
		medium estimate																
		upper bound																

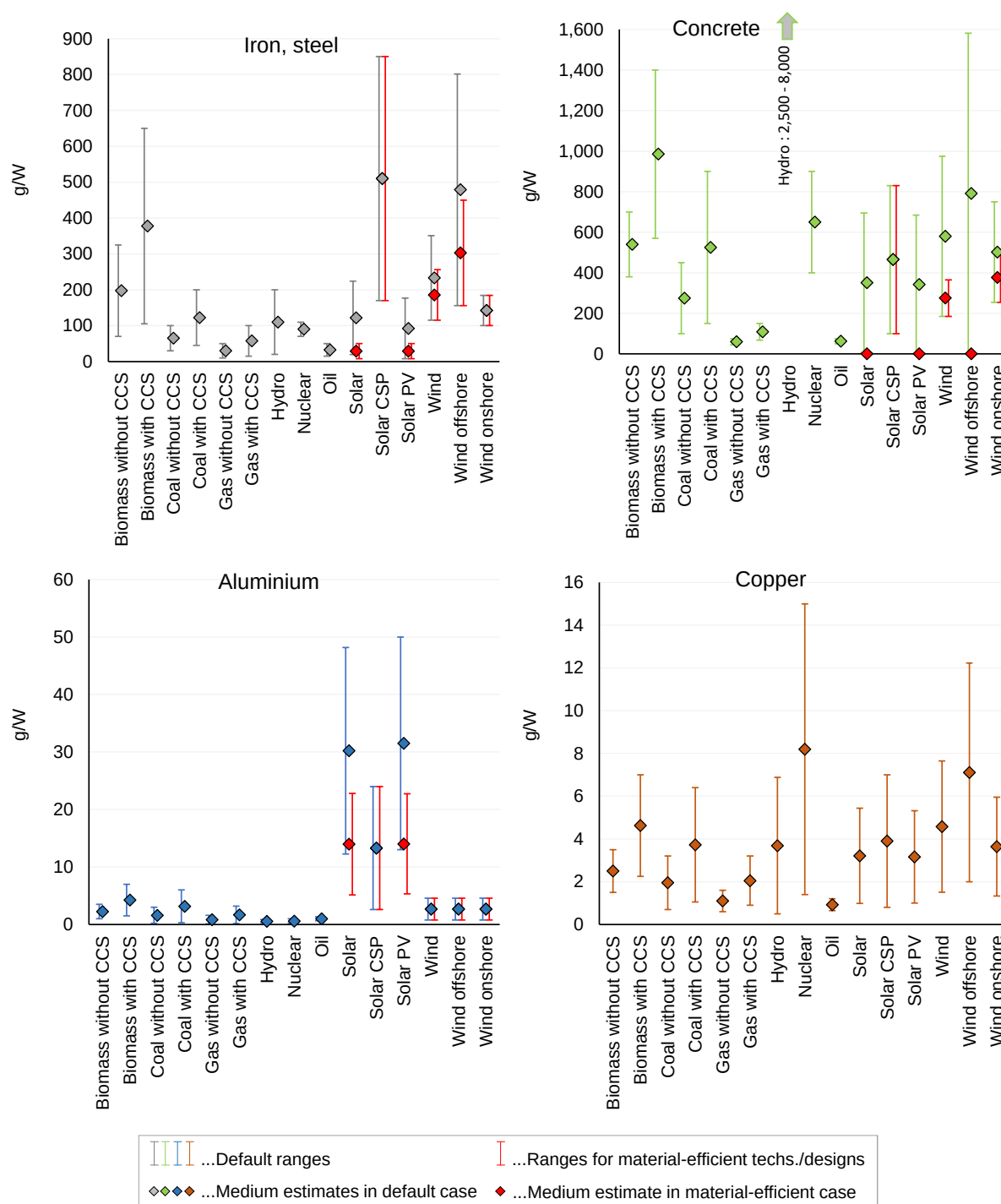


Figure S7. Material intensity ranges of electricity generation technologies (based on Kalt et al., 2021b).

2.2 Grids and transformers

The material intensity ranges of grid components are provided in Table S3. The data are adopted from Kalt et al. (2021b); please refer to this study for the underlying literature.

Different ranges for the material intensities per kilometre grid length among world regions are due to differences in voltage levels and average numbers of circuits per route (see Kalt et al., 2021a). We simplistically assume that these differences remain unchanged until 2050.

Table S3. Material intensity ranges for transmission and distribution grids and transformers (based on Kalt et al., 2021b)

Region →	Asian countries except Middle East, Japan and Former Soviet Union states		OECD 90 and EU member states and candidates		Middle East and Africa		Latin America and the Caribbean		Reforming Economies of Eastern Europe and the Former Soviet		All world regions		
Infrastructure component →	Trans-mission grid	Distri-bution grid	Trans-mission grid	Distri-bution grid	Trans-mission grid	Distri-bution grid	Trans-mission grid	Distri-bution grid	Trans-mission grid	Distri-bution grid	Trans-mission trans-formers	Distri-bution trans-formers	
Unit →	t/km circuit length												
Iron, steel	lower bound	18.42	1.97	15.03	1.42	15.00	1.97	22.01	1.97	15.07	1.63	420	2,000
	medium estimate	22.87	3.03	18.48	2.18	18.14	3.03	27.42	3.03	19.23	2.51	660	2,500
	upper bound	27.33	4.09	21.93	2.94	21.27	4.09	32.83	4.09	23.40	3.39	900	3,000
Concrete	lower bound	54.21	5.95	39.67	4.28	32.69	5.95	73.17	5.95	53.64	4.93	0	0
	medium estimate	71.87	7.43	53.74	5.35	43.90	7.43	100.22	7.43	72.57	6.17	0	0
	upper bound	89.54	8.92	67.80	6.41	55.10	8.92	127.26	8.92	91.50	7.40	0	0
Aluminium	lower bound	9.98	0.37	9.87	0.49	10.54	0.37	15.28	0.37	8.52	0.44	90	250
	medium estimate	15.70	0.74	15.57	0.80	16.66	0.74	23.31	0.74	13.68	0.78	100	350
	upper bound	21.43	1.11	21.26	1.11	22.78	1.11	31.33	1.11	18.83	1.11	110	450
Copper	lower bound	0.014	0.880	0.221	1.288	0.000	0.880	0.017	0.880	0.000	1.127	50	300
	medium estimate	0.017	1.885	0.274	2.385	0.000	1.885	0.020	1.885	0.000	2.188	100	350
	upper bound	0.020	2.891	0.326	3.482	0.000	2.891	0.024	2.891	0.000	3.249	150	400

2.3 Lifetimes of infrastructure components

The assumed lifetimes of infrastructure components, summarised in Table S4, are adopted from Kalt et al. (2021b). For hydropower plants, we differentiate between lifetime of machinery (i.e. metals) and dams (i.e. concrete).

Table S4. Assumed operational lifetimes of infrastructure components (see Kalt et al., 2021b).

Infrastructure		years
Power plants	Hydro (run-of-river)	50 / 150 ^a
	Hydro (storage)	50 / 150 ^a
	Geothermal	30
	Wind (onshore/offshore)	25
	Solar (PV/CSP)	25
	Bioenergy w/wo. CCS	20
	Coal w/wo. CCS	40
	Gas w/wo. CCS	30
	Oil	25
	Nuclear	50
Grids	Transmission	40
	Distribution	40
Transformers	Transmission transformers	35
	Distribution transformers	35

Comment:

a) 50 years for metals, 150 years for concrete (dams; see Hache et al., 2020)

3 Greenhouse gas intensities of bulk materials

We derive three projections for GHG intensities of bulk materials. Rationales and general assumptions for each projection as well as material-specific assumptions and references are summarized in Table S5 and explained in the following sub-sections.

3.1 Projection a: ‘Baseline’

For iron/steel and aluminium, we apply linear trend extrapolation to historical trends according to Wang et al. (2021) and IAI (2020), respectively (see Figure S8 and Figure S9). For direct CO₂ emissions from cement production, recent trends indicate a slight increase in global average values (IEA, 2020a); we thus assume that the total (process- and energy-related) emissions remain constant at 818 kg CO₂e/t cement (corresponding to 117 kg CO₂e/t concrete) in the ‘Baseline’ case (see Allwood et al., 2010; He et al., 2019; Lord et al., 2017; Worrell et al., 2001). For copper, the progressing decline of average ore grades is expected to result in rising energy consumption per unit copper (Kuipers et al., 2018; Mudd et al., 2013; Norgate and Haque, 2010; Elshkaki et al., 2016). For the ‘Baseline’ projection, we assume that the effects of declining ore grades offset technological efficiency increases, resulting in constant GHG intensities.

‘Baseline’ projections are assumed as upper limits for future GHG intensities. Thus, if the approaches used to derive projections b or c result in higher GHG intensities (as is the case for iron/steel and

aluminium in the High Coal scenario), the corresponding value from the ‘Baseline’ projection is used instead.

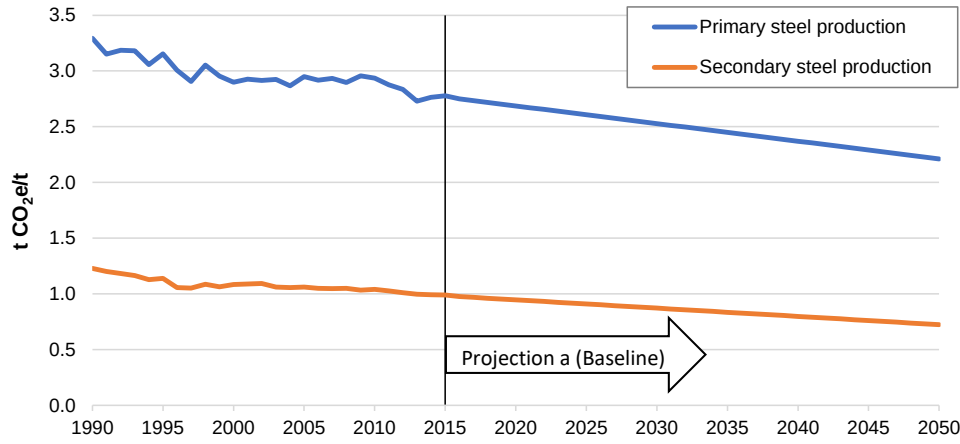


Figure S8. Baseline projection for the greenhouse gas intensity of iron/steel (linear extrapolation of historical trends according to Wang et al., 2021).

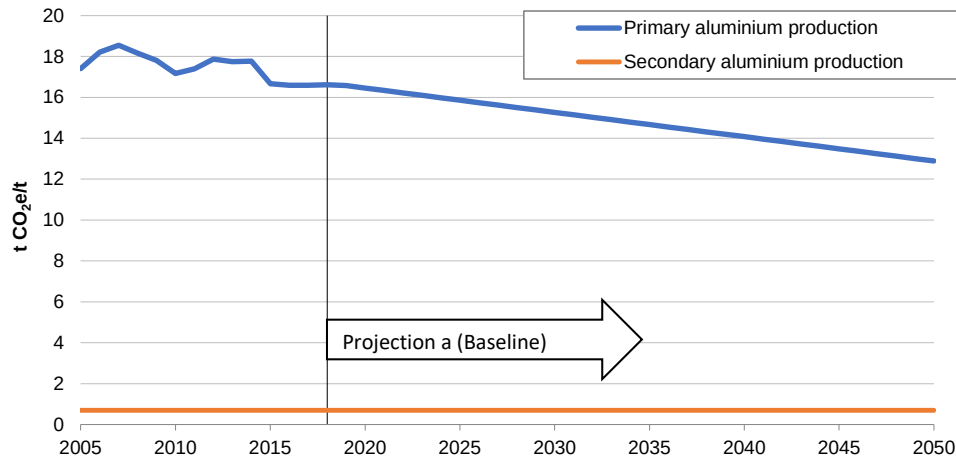


Figure S9. Baseline projection for the greenhouse gas intensity of aluminium (linear extrapolation of historical trends according to IAI, 2020).

3.2 Projection b: ‘Changing Energy mix’

In the projections b and c, the GHG intensities of materials ($GI_t^{mtrl,scen}$) are calculated as:

$$GI_t^{mtrl,scen} = \alpha_t^{scen} (s_t^{mtrl,prim} \cdot EE_{2021}^{mtrl,prim} + s_t^{mtrl,sec} \cdot EE_{2021}^{mtrl,sec}) + s_t^{mtrl,prim} \cdot PE_t^{mtrl,prim} \quad (1)$$

$EE_{2021}^{mtrl,prim}$ and $EE_{2021}^{mtrl,sec}$ represent the energy-related GHG emissions of primary and secondary material processing, respectively, and $s_t^{mtrl,prim}$ and $s_t^{mtrl,sec}$ the shares of primary/secondary material. These shares are assumed equal in all energy scenario, as material outflows from the

electricity sector only have little influence on the overall availability secondary materials. The default assumptions for primary and secondary material shares are presented below (section 3.4).

$PE_t^{mtrl,prim}$ denotes process-related GHG emissions of primary material processing. In projection b, processes emissions remain constant, i.e. $PE_t^{mtrl,prim} = PE_{2021}^{mtrl,prim}$. The assumed parameter values and material-specific assumptions and references are available from Table S5.

α_t^{scen} is a dimensionless scenarios-dependent parameter representing the change in average GHG intensity of final energy use, calculated as

$$\alpha_t^{scen} = \frac{EMI_t^{scen}/FE_t^{scen}}{EMI_{2021}^{scen}/FE_{2021}^{scen}}, \quad (2)$$

with EMI_t^{scen} denoting the total GHG emissions from the energy sector and FE_t^{scen} the final energy use. Figure S10 illustrates the development of this parameter in the five scenarios.

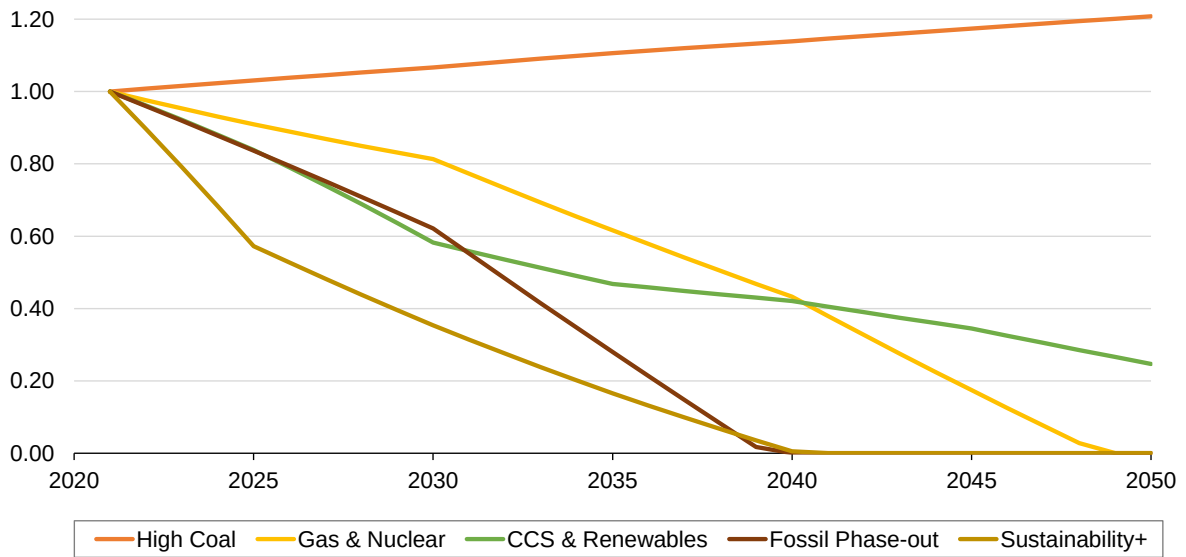


Figure S10. Relative development of greenhouse gas intensity of final energy in the five selected scenarios (Parameter α in Equation 1).

3.3 Projection c: ‘Transition to low-carbon processes’

GHG intensities for projection c are also calculated with the equations presented in the previous section. In contrast to projection b, the parameter $PE_t^{mtrl,prim}$ is time-dependent. The material specific assumptions, historical and assumed future parameter values are summarized in Table S5. Reductions in process emissions of the cement and aluminium industries imply technological innovations like carbon capture, utilization and storage; the specific reductions are adopted from IEA (2020b) and IAI (2021), respectively. Values for intermediate years are obtained through linear interpolation.

Table S5. Projections for greenhouse gas intensities of bulk materials: material-specific assumptions and references.

Projection	Rationale & general assumptions	Material-specific assumptions & references			
		Iron, steel	Cement/concrete	Aluminium	Copper
a) Baseline	GHG intensities follow current trends (linear projections in case of clear historical trends; constant intensities if this is not the case)	Linear extrapolation of historical data (Wang et al., 2021); 2015: 2.78 t CO₂e/t for primary and 0.99 t CO₂e/t for secondary steel; 2050: 2.21 and 0.72 t CO₂e/t , respectively	Constant at 818 kg CO₂e/t cement (based on (Allwood et al., 2010; Barcelo et al., 2014; He et al., 2019; Lord et al., 2017; Worrell et al., 2001)	Linear extrapolation of historical data (IAI, 2020); 2018: 16.61 t CO₂e/t for primary aluminium; 2050: 12.89 t CO₂e/t ; secondary aluminium constant at 0.7 t CO₂e/t	Constant at 6.3 t CO₂e/t for primary and 1.6 t CO₂e/t secondary copper (based on Dong et al., 2020; Kuipers et al., 2018; Van der Voet et al., 2019; Watari et al., 2020) ^a
b) Changing energy mix	Energy-related GHG emissions of processes decline at the same rate as emission intensity of final energy in the respective energy scenario; no shift to new (low-carbon) processes	Blast furnace remains dominant technology for ironmaking with process emissions at 1.45 t CO₂e/t (based on Carpenter, 2012; Rammer et al., 2017)	Process emissions constant at 450 kg CO₂e/t cement (see references above)	Direct emissions (anode consumption, non-CO ₂ -emission) constant at 2.49 t CO₂e/t aluminium (based on Huglen and Kvande (2016); IAI (2021, 2018, 2017, 2013)	No process GHG emissions; thus relative change of GHG intensity corresponds to relative change of final energy GHG intensity in the respective energy scenario
c) Transition to low-carbon processes	In addition to changing energy mix (b), transition to technologies with no/reduced direct/process emissions	Process emissions are eliminated through direct reduction with hydrogen; eventual differences in specific energy requirements of blast furnaces and direct reduction plants are disregarded	Average process emissions develop according to SDS scenario (IEA, 2020b): 200 kg CO₂e/t cement in 2050 ^b	Reduction of direct GHG emissions by 50% until 2050 (assumption based on IAI, 2021)	No process emissions, thus same as b)
Comments:					
a) Energy efficiency improvements are assumed to be offset by decreasing average ore grades, while the shares of pyrometallurgical and hydrometallurgical processing are assumed to remain constant.					
b) Reductions in cement process emissions are due to carbon capture, utilization and storage as well as a fall in the clinker-to-cement ratio (IEA, 2020b)-					

3.4 Shares of primary and secondary material

The development of primary and secondary raw material shares assumed for calculating GHG intensities is adopted from Watari and Yokoi (2021) (see Figure S11). In case of iron/steel and aluminium, the share of secondary material rises to about 50 % until 2050, and in case of copper to 37.5 %.

The relevance of assuming such rising secondary material shares in terms of GHG emissions is investigated in a sensitivity analysis in section 4.3 of this document, where constant shares of secondary material are assumed as alternative scenario variant.

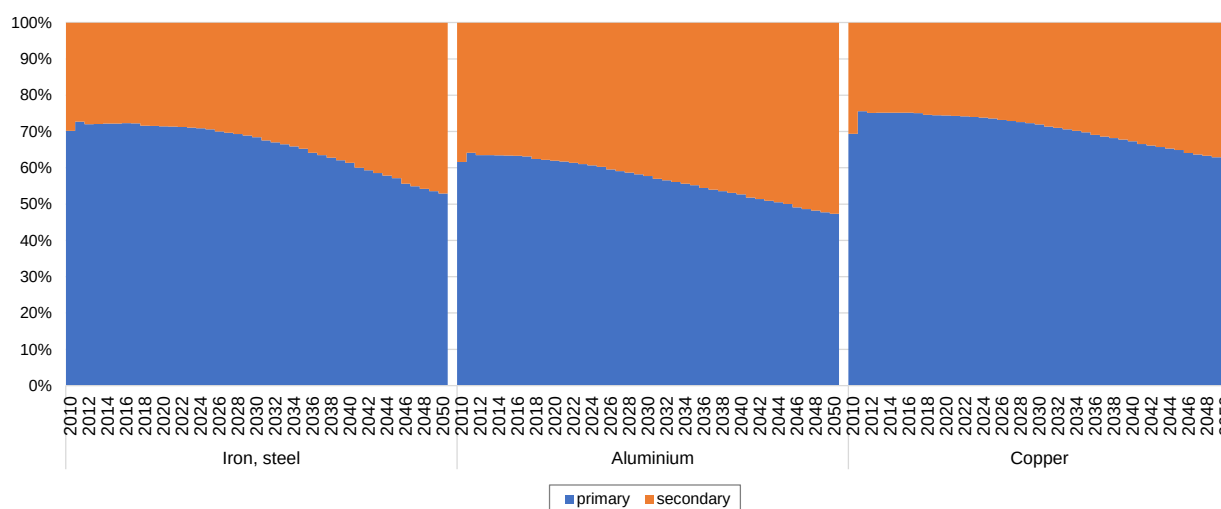


Figure S11. Default assumptions on the composition of metals supply by primary and secondary material (adopted from Watari and Yokoi, 2021).

3.5 GHG intensities of bulk materials

Table S6 provides the numerical values to the results on GHG intensities presented in Fig. 7a of the main article.

The use of secondary concrete as aggregates in concrete structures of the electricity sector is disregarded, thus only data for the case of constant (i.e. zero) share of secondary material are provided for concrete.

Table S6. Resulting greenhouse gas intensities of bulk materials, assuming increasing or constant shares of secondary material (increasing = default case, constant = sensitivity analysis; see section 4.3).

		2020	Projection a				Projection b				Projection c			
			2035		2050		2035		2050		2035		2050	
Share of secondary material:			const.	incr.	const.	incr.	const.	incr.	const.	incr.	const.	incr.	const.	incr.
Iron, steel	High Coal	2.189	1.988	1.888	1.786	1.511	1.988	1.888	1.786	1.511	1.794	1.729	1.393	1.328
	CCS & Renewables	2.189	1.988	1.888	1.786	1.511	1.576	1.478	1.320	1.038	1.058	1.005	0.285	0.271
	Fossil Phase-out	2.189	1.988	1.888	1.786	1.511	1.359	1.264	1.036	0.767	0.841	0.791	0.000	0.000
	Gas & Nuclear	2.189	1.988	1.888	1.786	1.511	1.747	1.646	1.036	0.767	1.229	1.173	0.000	0.000
	Sustainability+	2.189	1.988	1.888	1.786	1.511	1.227	1.134	1.036	0.767	0.709	0.661	0.000	0.000
Aluminium	High Coal	10.459	9.354	8.412	8.248	6.469	9.354	8.412	8.248	6.469	9.354	8.412	8.248	6.469
	CCS & Renewables	10.459	9.354	8.412	8.248	6.469	7.064	6.331	5.652	4.360	6.045	5.423	3.613	2.802
	Fossil Phase-out	10.459	9.354	8.412	8.248	6.469	5.865	5.248	4.077	3.116	4.845	4.339	2.039	1.558
	Gas & Nuclear	10.459	9.354	8.412	8.248	6.469	8.009	7.184	4.077	3.116	6.990	6.276	2.039	1.558
	Sustainability+	10.459	9.354	8.412	8.248	6.469	5.134	4.588	4.077	3.116	4.115	3.680	2.039	1.558
Copper	High Coal	5.097	5.097	4.879	5.097	4.539	5.097	4.879	5.097	4.539	5.097	4.879	5.097	4.539
	CCS & Renewables	5.097	5.097	4.879	5.097	4.539	2.385	2.284	1.258	1.120	2.385	2.284	1.258	1.120
	Fossil Phase-out	5.097	5.097	4.879	5.097	4.539	1.427	1.367	0.000	0.000	1.427	1.367	0.000	0.000
	Gas & Nuclear	5.097	5.097	4.879	5.097	4.539	3.140	3.006	0.000	0.000	3.140	3.006	0.000	0.000
	Sustainability+	5.097	5.097	4.879	5.097	4.539	0.844	0.808	0.000	0.000	0.844	0.808	0.000	0.000
Concrete	High Coal	0.117	0.117		0.117		0.117		0.117		0.105		0.092	
	CCS & Renewables	0.117	0.117		0.117		0.089		0.077		0.071		0.042	
	Fossil Phase-out	0.117	0.117	- ^a	0.117	- ^a	0.079	- ^a	0.064	- ^a	0.061	- ^a	0.029	- ^a
	Gas & Nuclear	0.117	0.117		0.117		0.097		0.064		0.079		0.029	
	Sustainability+	0.117	0.117		0.117		0.073		0.064		0.055		0.029	

Comments:

a) The share of secondary concrete is always assumed zero, therefore no values are provided for the case 'increasing share of secondary material'

4 Additional results

4.1 Metal requirements in the electricity sector as share of overall metal demand

The influence of giving preference to material-efficient solar and wind technologies/designs on the electricity sector's share in overall iron/steel and aluminium demand can be seen from Figure S12 (cf. Figure 6 in the main article). Apparently, wind and solar power have a considerable influence on the demand for these metals, and the uncertainty ranges are narrowed down significantly if material-efficient technologies/designs are assumed.

Copper is not considered here because no copper-sparing technologies/designs have been identified (see material intensities in section 2.1).

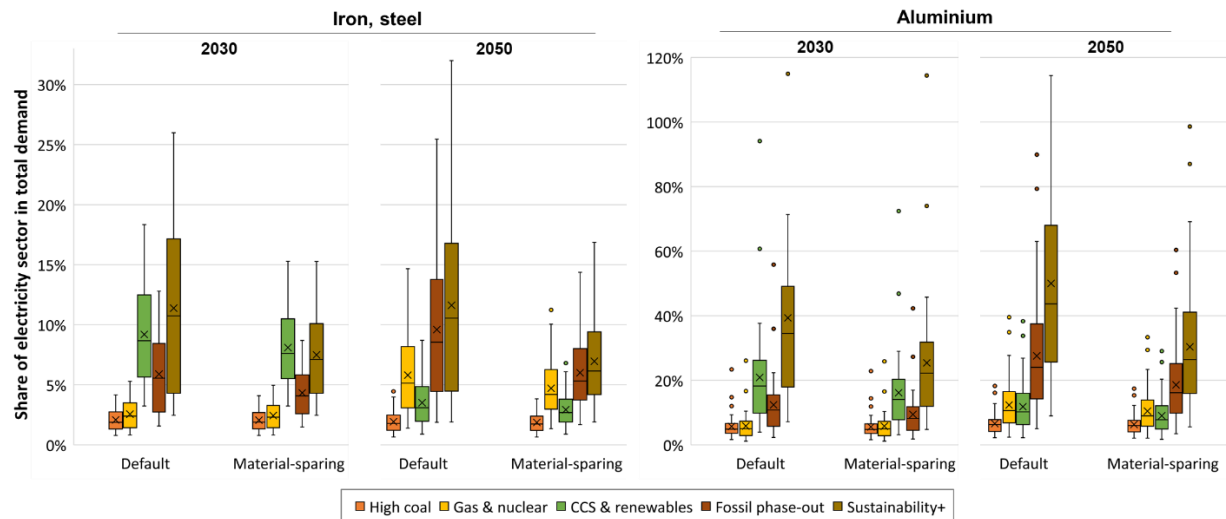


Figure S12. Global metal demand of electricity system as share of total metal demand according to projections in literature: Separate box plots for the default and ‘material-efficient’ case. Highest outlier for aluminium: 178 %.

4.2 Cumulative greenhouse gas emissions by material types

Figure S13 shows the composition of cumulative GHG emissions during 2021–2050 by bulk materials. The figure refers to the medium estimates (cf. Figure 7b of the main article); uncertainty bars for low and high estimates are omitted here.

Copper and concrete are generally clearly less relevant in terms of GHG emissions than iron/steel and aluminium. GHG emissions from copper processing are especially low in scenarios with rapid energy system decarbonisation because there are no process emissions involved.

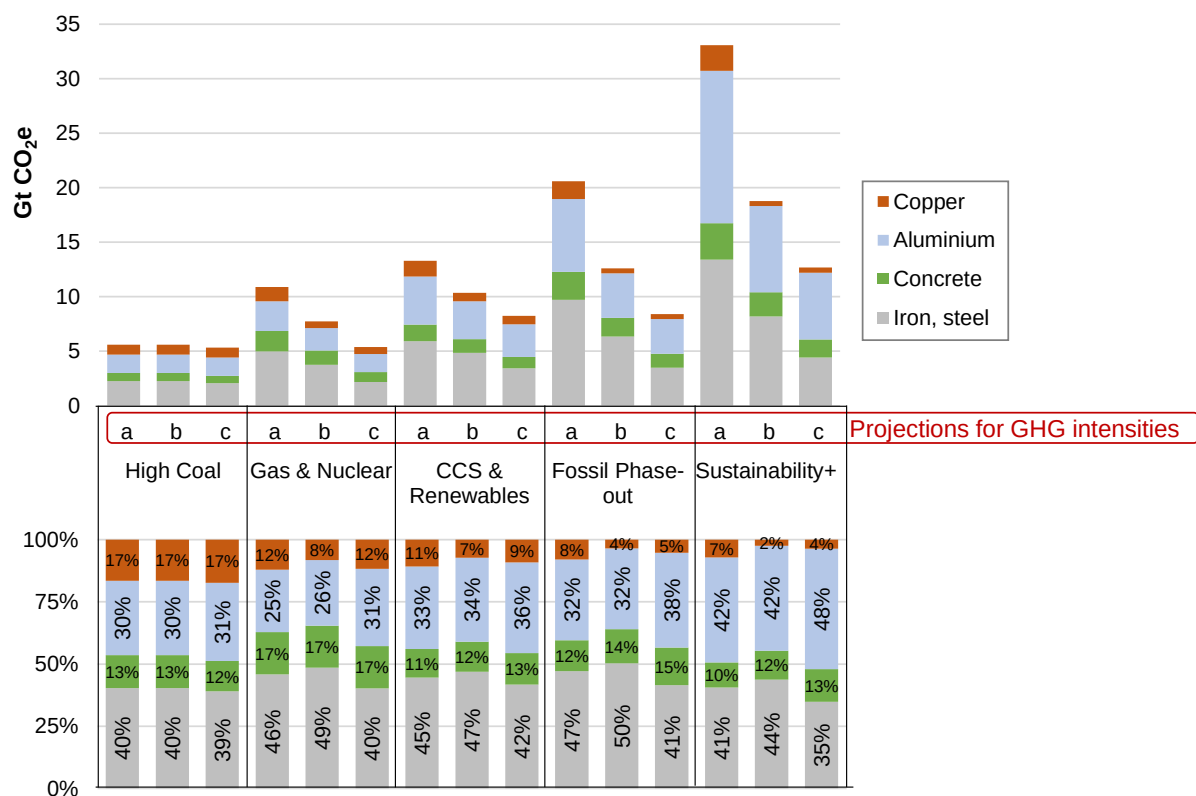


Figure S13. Cumulative GHG emissions (2021-2050) from bulk material consumption in the five exemplary scenarios: Medium estimates based on full range of material intensities for solar and wind.

4.3 Sensitivity analysis: Constant shares of secondary materials

The relevance of our default assumption of an increase in secondary metal usage (Figure S11) is investigated here. Figure S14 shows the difference in cumulative GHG emissions between projections based on constant and increasing secondary material share. If secondary material shares are assumed constant, the cumulative GHG emissions are about 5 to 11.5 % higher than in the default case shown in Figure 7b of the main article. This corresponds to about 0.2 to 5 Gt CO₂e.

Obviously, the GHG benefits from metal recycling strongly depend on the GHG intensities of primary material processing, which are significantly higher than those of secondary material processing for all considered metals. The higher the GHG intensity of primary material processing in relation to secondary material processing, the higher the GHG benefits from recycling. Note that GHG emissions from secondary material processing are 100 % energy-related (see Equ. 1), and are thus linked to the GHG intensity of final energy supply in the projections b and c. GHG emissions from secondary material processing are generally significantly smaller than those of primary material processing, especially for aluminium.

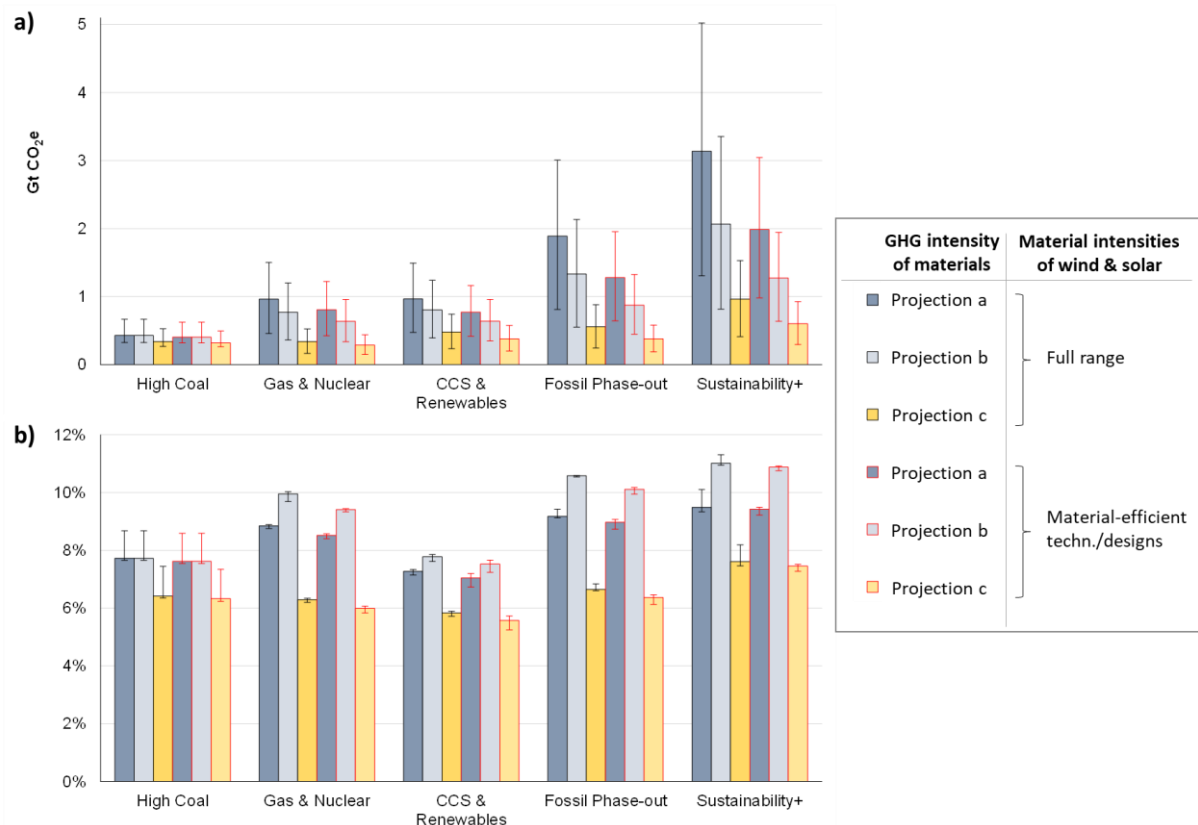


Figure S14. Sensitivity of cumulative GHG emissions (2021-2050) regarding the share of secondary materials: Difference between projections based on constant and increasing secondary material share in absolute (a) and relative numbers (b).

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